

MATH1231 CALCULUS

Chapter 3B

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Practice Problems – 1st Order ODEs

► $xy' - 2y = x^3 \quad y(1) = 0$

► $y' = \frac{e^y}{x^2 + 1} \quad y(0) = 1$

► $(x^2y + y^3)\frac{dy}{dx} + (xy^2 + \sinh 2x) = 0$

(3.7) Modelling with First Order ODEs

Derivatives express a rate of change of one variable w.r.t. another and they appear naturally in many modelling equations, e.g.,

bacterial growth $\frac{dN}{dt} = kN$

drug decay $\frac{dy}{dt} = -ky$

motion $\frac{d(mv)}{dt} = F$

cooling $\frac{dT}{dt} = -k(T(t) - T_a)$

mixing $\frac{dV}{dt} = V_{in} - V_{out}$

(3.7) Modelling Example

November 1997 Final Exam

A student carrying a flu virus returns to an isolated college campus. The rate at which the virus spreads is proportional to the product of the number x of students infected and the number not infected.

- (a) Assuming that the total number of students on campus is N and no one leaves the campus during the disease write down a formula for the rate of change of x with respect to time. What is the initial condition?
- (b) If $N = 1000$ and the number infected after 4 days is 100 how many will be infected after 8 days?

Modelling Example

Newton's Law of Cooling

The rate of decrease of temperature is proportional to the difference between the temperature of an object and its surroundings

$$\frac{dT}{dt} = -k(T(t) - T_A) \quad \left\{ \begin{array}{l} T_A \text{ ambient temp. in } ^\circ\text{C} \\ T(t) \text{ temp. of object after time } t \\ k \text{ unknown proportionality constant (inverse time)} \end{array} \right.$$

This is widely used in forensics.

A dead body is discovered by a cleaner in a hotel room at 11am. A forensic scientist arrives at the scene at 11.30am and records the temperature of the body at 24.5°C . The body temperature is recorded again an hour later at 24.0°C . Estimate the time of death given that the hotel room is at constant temp. 20°C and normal body temp. is 36.5°C .

Modelling Example: World Population

Let $N(t)$ denote the population at time t :

$$\frac{dN}{dt} = g(N) \quad \text{growth rate}$$

$$\frac{1}{N} \frac{dN}{dt} = \frac{1}{N} g(N) \quad \text{per capita growth rate}$$

Note

$$g(N) > 0 \quad \Rightarrow \quad \text{birth rate exceeds death rate}$$

$$g(N^*) = 0 \quad \Rightarrow \quad \text{birth rate equals death rate}$$

$$g(0) = 0 \quad \Rightarrow \quad \text{no births or deaths}$$

How can we find the functional form of $g(N)$?

Growth Model

Assume that in a given time interval a fixed percentage of the population is born and a fixed percentage dies then

$$N(t + \Delta t) = N(t) + bN(t)\Delta t - cN(t)\Delta t$$

b percentage that is born per unit time

c percentage that dies per unit time

$$\begin{aligned} \frac{N(t + \Delta t) - N(t)}{\Delta t} &= (b - c)N(t) \\ \Rightarrow \frac{dN}{dt} &= kN \quad \text{constant per capita growth} \end{aligned}$$

Linear Per Capita Growth

Assume per capita growth declines linearly as N approaches a threshold population level N^* .

$$k = r \left(1 - \frac{N}{N^*} \right)$$

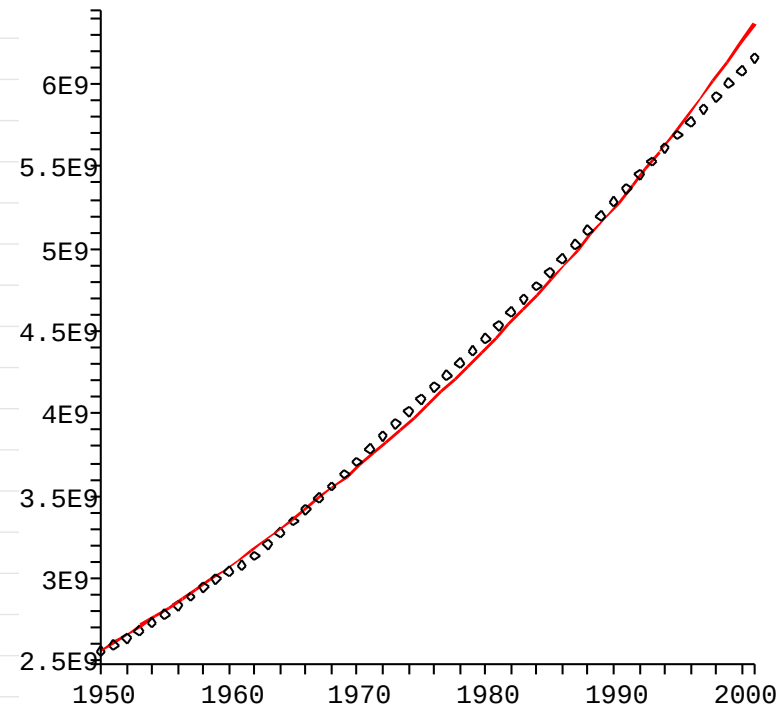
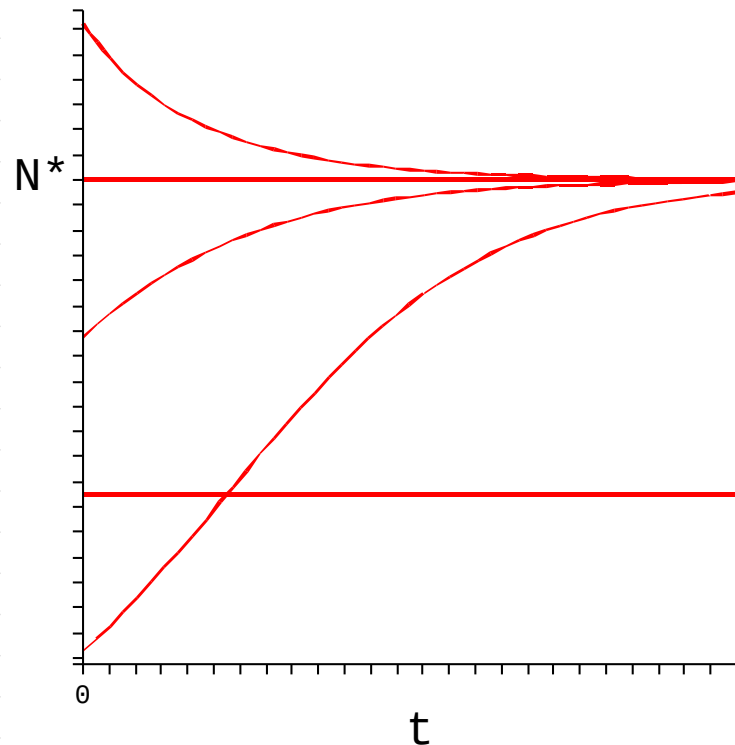
k and r have the same units hence r is the percentage growth per unit time

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{N^*} \right)$$

This first order ODE is separable. EX. Find the general solution.

The world population is currently approx. 7.8 billion. The UN estimates it will grow to approximately 12 billion.

Left: Solutions for linear per capita growth – population reaches a threshold level



Right: Real world data compared with solution for constant per capita growth – is the real data approaching a threshold?

Mixing

At time $t = 0$ a tank contains 1 kg of salt dissolved in 100 liters of water. If salty water containing $1/4$ kg per litre flows into the tank at a rate of 3 litres per minute and the (stirred, and hence well-mixed) solution is draining from the tank at 3 litres per minute, predict how much salt will be in the tank at time t .



BASE Jumping

Part of the thrill in BASE jumping is that high free fall speeds can be achieved in short jumps. The speed v satisfies the ODE

$$m \frac{dv}{dt} = mg - cv^2 \quad v(0) = 0$$

m mass of jumper, g acceln due to gravity, c air resistance const.

- (i) Find an expression for the terminal speed of a BASE jumper. (The terminal speed has been measured at $\approx 53 \text{ ms}^{-1}$.)
- (ii) How long does it take to reach 95% terminal speed?
- (iii) What starting height is needed to enable the BASE jumper to reach 95% terminal speed?